

Lab 01

1. Establish a local repository for your project with a descriptive name. Set up Git configurations by adding a username and email address. Display and modify the Git configurations, and verify if the changes are reflected in the .git/config file post-editing. Showcase the utilization of both Git Local and Global configurations.

Step 1 :- Establish local repository with descriptive name

```
avej_lanjekar@AvejLanjekar:~$ mkdir git-lab
avej_lanjekar@AvejLanjekar:~$ cd git-lab/
avej_lanjekar@AvejLanjekar:~/git-lab$ ls
avej_lanjekar@AvejLanjekar:~/git-lab$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:     git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:     git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/avej_lanjekar/git-lab/.git/
avej_lanjekar@AvejLanjekar:~/git-lab$ |
```

Step 2 :- Set up Git configurations by adding a username and email address.

```
avej_lanjekar@AvejLanjekar:~/git-lab$ git config --global user.name "Avej K Lanjekar"
avej_lanjekar@AvejLanjekar:~/git-lab$ git config --global user.email "avej.lanjekar11@gmail.com"
avej_lanjekar@AvejLanjekar:~/git-lab$ |
```

Step 3:- Display and modify the Git configurations, and verify if the changes are reflected in the .git/config

```
avej_lanjekar@AvejLanjekar:~/git-lab$
avej_lanjekar@AvejLanjekar:~/git-lab$ git config --global -l
user.name=Avej K Lanjekar
user.email=avej.lanjekar11@gmail.com
avej_lanjekar@AvejLanjekar:~/git-lab$ |
```

```
avej_lanjekar@AvejLanjekar:~/git-lab$ git config -l
user.name=Avej K Lanjekar
user.email=avej.lanjekar11@gmail.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
user.name=Mahammadavej
user.email=avejlanjekar786@gmail.com
avej_lanjekar@AvejLanjekar:~/git-lab$ |
```

```

avej_lanjekar@AvejLanjekar:~/git-lab$ git config user.name Mahammadavej
avej_lanjekar@AvejLanjekar:~/git-lab$ git config user.email avejlanjekar786@gmail.com
avej_lanjekar@AvejLanjekar:~/git-lab$ ls -la
total 12
drwxr-xr-x 3 avej_lanjekar avej_lanjekar 4096 Jul 28 06:52 .
drwxr-xr-x 8 avej_lanjekar avej_lanjekar 4096 Jul 28 07:11 ..
drwxr-xr-x 6 avej_lanjekar avej_lanjekar 4096 Jul 28 07:12 .git
avej_lanjekar@AvejLanjekar:~/git-lab$ cd .git
avej_lanjekar@AvejLanjekar:~/git-lab/.git$ ls
HEAD  config  description  hooks  info  objects  refs
avej_lanjekar@AvejLanjekar:~/git-lab/.git$ cat config
[core]
    repositoryformatversion = 0
    filemode = true
    bare = false
    logallrefupdates = true
[user]
    name = Mahammadavej
    email = avejlanjekar786@gmail.com
avej_lanjekar@AvejLanjekar:~/git-lab/.git$ |

```

2. Set up a local repository for your project with a meaningful name. Explore the default branch in the local Git repository. Sequentially create five files, committing each with informative messages. Share the Git log output through multiple methods once the process is complete.

Step 1:- create local Repository

```

avej_lanjekar@AvejLanjekar:~$ mkdir git-practice
avej_lanjekar@AvejLanjekar:~$ cd git-practice/
avej_lanjekar@AvejLanjekar:~/git-practice$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:     git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:     git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/avej_lanjekar/git-practice/.git/

```

Step 2:- Sequentially create five files

```

avej_lanjekar@AvejLanjekar:~/git-practice$ echo "Hello cloudEthix , file 1" > file1.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ echo "Hello cloudEthix , file 2" > file2.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ echo "Hello cloudEthix , file 3" > file3.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ echo "Hello cloudEthix , file 4" > file4.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ echo "Hello cloudEthix , file 5" > file5.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ ls
file1.txt  file2.txt  file3.txt  file4.txt  file5.txt

```

Step 3:- committing each with informative messages

```
avej_lanjekar@AvejLanjekar:~/git-practice$ git add file1.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ status
Command 'status' not found, did you mean:
  command 'statup' from snap statup (0.79.91)
  command 'statfs' from deb gocryptfs (2.6.1-1)
  command 'states' from deb enscrip (1.6.5.90-3.2build1)
  command 'qstatus' from deb gridengine-client (8.1.9+dfsg-13.2)
See 'snap info <snapname>' for additional versions.
avej_lanjekar@AvejLanjekar:~/git-practice$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   file1.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file2.txt
        file3.txt
        file4.txt
        file5.txt

avej_lanjekar@AvejLanjekar:~/git-practice$ git commit -m "Added first file"
[master (root-commit) a2f3c6b] Added first file
 1 file changed, 1 insertion(+)
 create mode 100644 file1.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ git add file2.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ git commit -m "Added Second file"
[master edee63d] Added Second file
 1 file changed, 1 insertion(+)
 create mode 100644 file2.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file3.txt
        file4.txt
        file5.txt
```

```
avej_lanjekar@AvejLanjekar:~/git-practice$ git add file3.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ git commit -m "Added Third file"
[master 67e02e8] Added Third file
 1 file changed, 1 insertion(+)
 create mode 100644 file3.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ git add file4.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ git commit -m "Added Fourth file"
[master eb3df77] Added Fourth file
 1 file changed, 1 insertion(+)
 create mode 100644 file4.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ git add file5.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ git commit -m "Added Fifth file"
[master 75332a2] Added Fifth file
 1 file changed, 1 insertion(+)
 create mode 100644 file5.txt
avej_lanjekar@AvejLanjekar:~/git-practice$ git status
On branch master
nothing to commit, working tree clean
avej_lanjekar@AvejLanjekar:~/git-practice$ |
```

Step 4:- Git log output through multiple methods

Command:- Git log

```
avej_lanjekar@AvejLanjekar:~/git-practice$ git log
commit 75332a284e8190683ebdf04cf815fd2b0a878365 (HEAD -> master)
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date: Tue Jul 28 07:32:22 2026 +0000

    Added Fifth file

commit eb3df7720000672262ccd9f7109e49e2cafbf4bf
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date: Tue Jul 28 07:32:09 2026 +0000

    Added Fourth file

commit 67e02e86ec745c82dbeadb3841620fd3d637476b
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date: Tue Jul 28 07:31:56 2026 +0000

    Added Third file

commit edee63d4e5cbe1513c6a999daa059439c59aedef3
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date: Tue Jul 28 07:30:48 2026 +0000

    Added Second file

commit a2f3c6b633c262d5a40764b041c27c94f9c4c527
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date: Tue Jul 28 07:30:25 2026 +0000

    Added first file
avej_lanjekar@AvejLanjekar:~/git-practice$ |
```

Command:- git log --oneline

```
avej_lanjekar@AvejLanjekar:~/git-practice$ git log --oneline
75332a2 (HEAD -> master) Added Fifth file
eb3df77 Added Fourth file
67e02e8 Added Third file
edee63d Added Second file
a2f3c6b Added first file
avej_lanjekar@AvejLanjekar:~/git-practice$ |
```

Command:- git shortlog

```
avej_lanjekar@AvejLanjekar:~/git-practice$ git shortlog
Avej K Lanjekar (5):
    Added first file
    Added Second file
    Added Third file
    Added Fourth file
    Added Fifth file
```

Command:- git log --stat

```
avej_lanjekar@AvejLanjekar:~/git-practice$ git log --stat
commit 75332a284e8190683ebdf04cf815fd2b0a878365 (HEAD -> master)
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date: Tue Jul 28 07:32:22 2026 +0000

    Added Fifth file

file5.txt | 1 +
1 file changed, 1 insertion(+)

commit eb3df7720000672262ccd9f7109e49e2cafbf4bf
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date: Tue Jul 28 07:32:09 2026 +0000

    Added Fourth file

file4.txt | 1 +
1 file changed, 1 insertion(+)

commit 67e02e86ec745c82dbeadb3841620fd3d637476b
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date: Tue Jul 28 07:31:56 2026 +0000

    Added Third file

file3.txt | 1 +
1 file changed, 1 insertion(+)

commit edee63d4e5cbe1513c6a999daa059439c59aedef3
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date: Tue Jul 28 07:30:48 2026 +0000

    Added Second file

file2.txt | 1 +
1 file changed, 1 insertion(+)

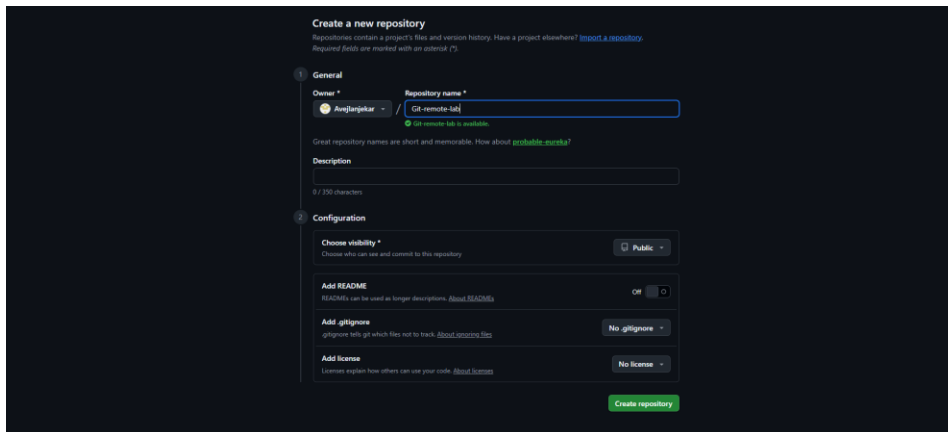
commit a2f3c6b633c262d5a40764b041c27c94f9c4c527
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date: Tue Jul 28 07:30:25 2026 +0000
```

Command:- git log --oneline --graph

```
avej_lanjekar@AvejLanjekar:~/git-practice$ git log --oneline --graph
* 75332a2 (HEAD -> master) Added Fifth file
* eb3df77 Added Fourth file
* 67e02e8 Added Third file
* edee63d Added Second file
* a2f3c6b Added first file
```

3. Open a GitHub account and initiate a remote Git repository. Examine the default branch in the remote repository. Create and commit the following files one by one with meaningful commit messages:
 - i. README.md
 - ii. send_disk_alert.sh
 - iii. create_users.sh
 - iv. check_service.sh.

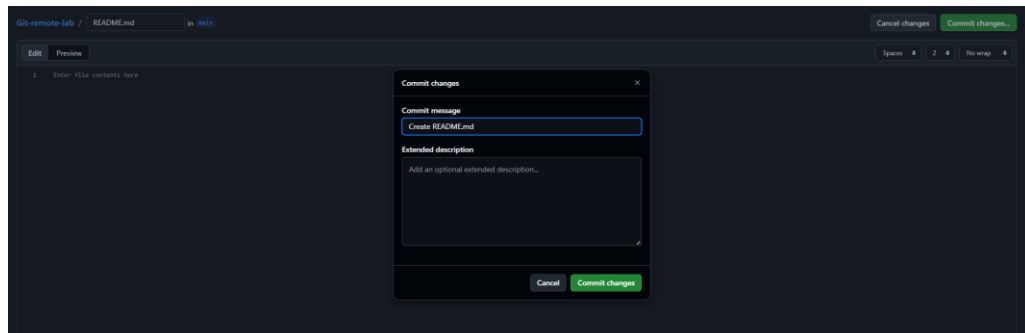
step 1:- Open a GitHub account and initiate a remote Git repository



The screenshot shows the 'Create a new repository' page on GitHub. It has two main sections: 'General' and 'Configuration'. In the 'General' section, the 'Repository name' is 'git-remote-lab' and the 'Description' is empty. In the 'Configuration' section, 'Choose visibility' is set to 'Public', 'Add README' is checked, 'Add .gitignore' is set to 'No .gitignore', and 'Add license' is set to 'No license'. A green 'Create repository' button is at the bottom right.

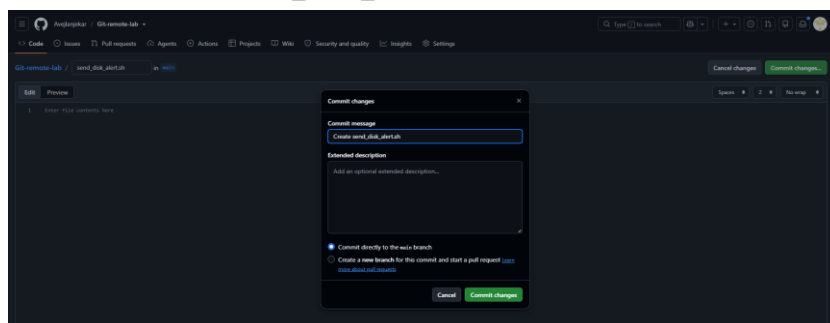
Step 2:- Create and commit the following files one by one with meaningful commit messages

a) Commit the File Readme.md



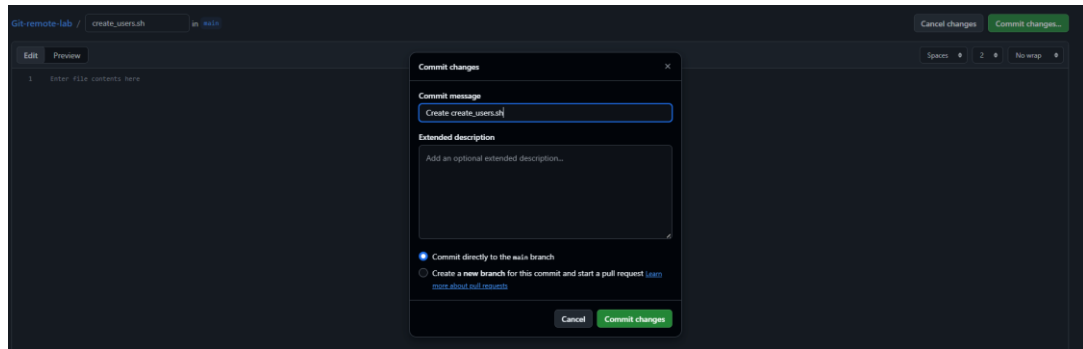
The screenshot shows the 'Commit changes' dialog box in GitHub. The 'Commit message' field contains 'Create README.md'. The 'Extended description' field is empty. There are 'Cancel' and 'Commit changes' buttons at the bottom.

b) Commit the File send_disk_alert.sh

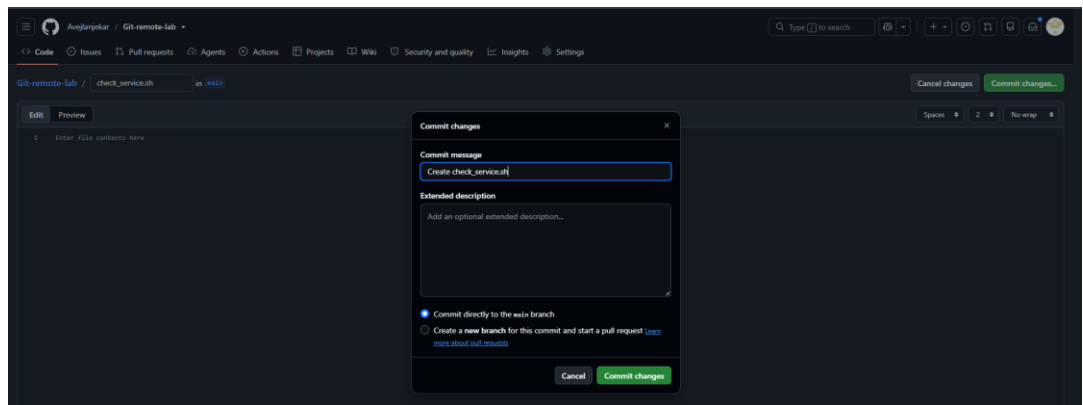


The screenshot shows the 'Commit changes' dialog box in GitHub. The 'Commit message' field contains 'Create send_disk_alert.sh'. The 'Extended description' field is empty. At the bottom, there are two radio buttons: 'Commit directly to the main branch' (which is selected) and 'Create a new branch for this commit and start a pull request'. There are 'Cancel' and 'Commit changes' buttons at the bottom.

c) Commit the File create_users.sh



d) Commit the file check_service.sh



4. Establish a local repository for your project with a descriptive name. Illustrate the implementation of a Git-Ignore file by committing multiple files to the local repository.

Step1:- Establish a local repository for your project with a descriptive name

```
avej_lanjekar@AvejLanjekar:~$ mkdir git_ignore_workshop
avej_lanjekar@AvejLanjekar:~$ cd git_ignore_workshop/
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ ls
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/avej_lanjekar/git_ignore_workshop/.git/
```

Step 2 :- Implementation of git-ignore file by commenting multiple files to local repo

```
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ touch server.js index.html style.css config.env node_modules
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ ls
config.env index.html node_modules server.js style.css
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ git status
On branch master
```

No commits yet

```
Untracked files:
(use "git add <file>..." to include in what will be committed)
config.env
index.html
node_modules
server.js
style.css
```

nothing added to commit but untracked files present (use "git add" to track)

```
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ vim .gitignore
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ git status
On branch master
```

No commits yet

```
Untracked files:
(use "git add <file>..." to include in what will be committed)
.gitignore
index.html
server.js
style.css
```

nothing added to commit but untracked files present (use "git add" to track)

```
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ git add .
```

```
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ git status
On branch master
```

No commits yet

```
Changes to be committed:
(use "git rm --cached <file>..." to unstage)
new file:   .gitignore
new file:   index.html
new file:   server.js
new file:   style.css
```

```
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ git commit -m "Added files"
[master (root-commit) aa2fc12] Added files
4 files changed, 2 insertions(+)
create mode 100644 .gitignore
create mode 100644 index.html
create mode 100644 server.js
create mode 100644 style.css
```

```
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ git status
On branch master
```

nothing to commit, working tree clean

```
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ git status --ignored
```

On branch master

```
Ignored files:
(use "git add -f <file>..." to include in what will be committed)
config.env
node_modules
```

nothing to commit, working tree clean

```
avej_lanjekar@AvejLanjekar:~/git_ignore_workshop$ |
```


5. Create a local repository with a meaningful project name. Demonstrate the utilization of GIT working area, staging area, and commit area. Experiment with committing, deleting, and restoring files from the repository. Additionally, add files to the staging area and restore them from the staging area.

Step 1:- Create a local repository with a meaningful project name

```
avej_lanjekar@AvejLanjekar:~/git-workbench$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:     git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:     git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/avej_lanjekar/git-workbench/.git/
```

Step 2:- Demonstrate the utilization of GIT working area, staging area, and commit area.

```
avej_lanjekar@AvejLanjekar:~/git-workbench$ echo "Hello Cloudethix team" > app.txt
avej_lanjekar@AvejLanjekar:~/git-workbench$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  app.txt

nothing added to commit but untracked files present (use "git add" to track)
avej_lanjekar@AvejLanjekar:~/git-workbench$ git add app.txt
avej_lanjekar@AvejLanjekar:~/git-workbench$ git commit -m "initial commit adding app.txt"
[master (root-commit) aafd2b4] initial commit adding app.txt
 1 file changed, 1 insertion(+)
 create mode 100644 app.txt
avej_lanjekar@AvejLanjekar:~/git-workbench$ git status
On branch master
nothing to commit, working tree clean
```

Step 3:- Experiment with committing, deleting, and restoring files from the repository

```
avej_lanjekar@AvejLanjekar:~/git-workbench$ git add app.txt
avej_lanjekar@AvejLanjekar:~/git-workbench$ git commit -m "initial commit adding app.txt"
[master (root-commit) aafd2b4] initial commit adding app.txt
 1 file changed, 1 insertion(+)
 create mode 100644 app.txt
avej_lanjekar@AvejLanjekar:~/git-workbench$ git status
On branch master
nothing to commit, working tree clean
avej_lanjekar@AvejLanjekar:~/git-workbench$ rm -f app.txt
avej_lanjekar@AvejLanjekar:~/git-workbench$ ls
avej_lanjekar@AvejLanjekar:~/git-workbench$ git restore app.txt
avej_lanjekar@AvejLanjekar:~/git-workbench$ ls
app.txt
avej_lanjekar@AvejLanjekar:~/git-workbench$ git status
On branch master
nothing to commit, working tree clean
avej_lanjekar@AvejLanjekar:~/git-workbench$ ls
app.txt
avej_lanjekar@AvejLanjekar:~/git-workbench$ cat app.txt
Hello Cloudethix team
```

Step 4:- Additionally, add files to the staging area and restore them from the staging area

```
avej_lanjekar@AvejLanjekar:~/git-workbench$ echo "Address:- Baner,pune Maharashtra" >> app.txt
avej_lanjekar@AvejLanjekar:~/git-workbench$ cat app.txt
Hello Cloudethix team
Address:- Baner,pune Maharashtra
avej_lanjekar@AvejLanjekar:~/git-workbench$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.txt

no changes added to commit (use "git add" and/or "git commit -a")
avej_lanjekar@AvejLanjekar:~/git-workbench$ git add app.txt
avej_lanjekar@AvejLanjekar:~/git-workbench$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   app.txt

avej_lanjekar@AvejLanjekar:~/git-workbench$ git restore --staged app.txt
avej_lanjekar@AvejLanjekar:~/git-workbench$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.txt

no changes added to commit (use "git add" and/or "git commit -a")
avej_lanjekar@AvejLanjekar:~/git-workbench$ |
```